

Numerical Methods for the Quantum Harmonic Oscillator and Hydrogen Atom

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22nd November 2025

We develop and evaluate numerical eigen-solvers for the 1D harmonic oscillator and the ($\ell = 0$) hydrogen radial problem, using numerical solvers whose parameters are tuned to meet specific accuracy and runtime goals [1]. Our investigation focuses on how choices such as grid spacing, domain size, and discretisation order determine the reliability of the spectra we obtain. To contextualise the results, we compare Numerov shooting with the finite-difference solver using parameter scans and error-scaling tests. All computations are checked against analytic energies to ensure that boundary and discretisation errors stay below the target tolerance.

1 Methods

1.1 Discretisation strategy

Both systems use atomic units and share the discrete Hamiltonian form

$$H = -\frac{1}{2} D^{(2)} + \text{diag}(V),$$

where $D^{(2)}$ is the second-derivative matrix and V is the potential; only V differs between cases.

Quantum harmonic oscillator (QHO). For $H = -\frac{1}{2} \partial_{xx} + \frac{1}{2} x^2$ with exact $E_n = n + \frac{1}{2}$, we discretise $x \in [-L, L]$ using Dirichlet boundaries and a centred three-point (2nd-order) Laplacian, producing a sparse tridiagonal matrix. A five-point stencil is used only once boundary effects are negligible. The box size L is set by checking that $|\psi(\pm L)|$ is effectively zero so that finite-difference error dominates.

Hydrogen ($\ell = 0$). For the radial operator $H = -\frac{1}{2} \partial_{rr} - \frac{1}{r}$ with $E_n = -1/(2n^2)$, we use $r \in [0, R_{\max}]$ and enforce $u(0) = u(R_{\max}) = 0$. Two schemes are compared: (i) three-point finite differences on the interior nodes (sparse tridiagonal), and (ii) a Numerov shooting method with node counting, suited to $u'' = f(r)u$. $R_{\max}$ is chosen from diagnostics (energy plateaus and $\langle r \rangle$ stability with N) to keep truncation errors subdominant.

1.2 Solvers and algorithmic considerations

After discretisation the Hamiltonians are real, symmetric, banded, and only the lowest few eigenpairs are required. Dense diagonalisation (`eigh`) is used as a high-accuracy reference but its $O(N^3)$ cost restricts it to small grids. All production results therefore use the sparse Lanczos solver (`eigsh`), whose cost is governed by matrix–vector products. With the three-point stencil the Hamiltonian

has $\text{nnz} \sim O(N)$, giving an almost linear $O(kN)$ cost for the lowest k states. A five-point stencil is tested for the QHO to assess whether higher FD order compensates for the added bandwidth; the asymptotic scaling is unchanged.

For hydrogen we also employ a Numerov shooting scheme. Unlike Lanczos, which acts directly on the discretised operator, Numerov integrates the underlying ODE via a sixth-order step and identifies eigenvalues by bisection guided by node counts. Its asymptotic regime offers higher accuracy and strictly linear runtime, but requires excellent resolution near $r = 0$ and sufficiently large $R_{\max}$. With moderate N these conditions may not be fully met, motivating comparison with the three-point FD discretisation. Since both arise from the same Sturm–Liouville operator, differences cleanly reflect the contrast between high-order ODE integration and sparse-matrix eigensolution [2].

1.3 Accuracy targets and validation

We impose a uniform accuracy requirement on both systems: the largest deviation among the computed eigenvalues must satisfy

$$\max_n |E_{n,\text{num}} - E_n^{\text{exact}}| \leq 10^{-3}.$$

Because higher states carry larger discretisation errors, this condition effectively forces even tighter accuracy on the low-lying levels. The tolerance is designed to ensure: qualitative structure such as node counts, offsets, and parity is not distorted; eigenvalues remain well separated relative to spectral gaps, avoiding ambiguity in their ordering; further increases in grid size would yield only marginal improvements relative to the added cost.

Validation proceeds by direct comparison with the analytic spectra, while log–log error trends against N are used to verify the expected convergence rates of the chosen discretisations. Here p refers to the finite-difference convergence order, i.e. the slope of the log–error versus log–resolution curve. The exponent α denotes the empirical runtime–scaling rate, obtained from log–log fits of CPU time against grid size.

2 Results and Discussion

2.1 Quantum harmonic oscillator (QHO)

2.1.1 Domain and grid parameter calibration

We choose $L = 8$ and $N = 1400$ as the smallest grid on which the highest state considered ($n = 9$) meets the tolerance $|E_9 - 9.5| < 10^{-3}$; the lower, more localised levels automatically satisfy this criterion. These values

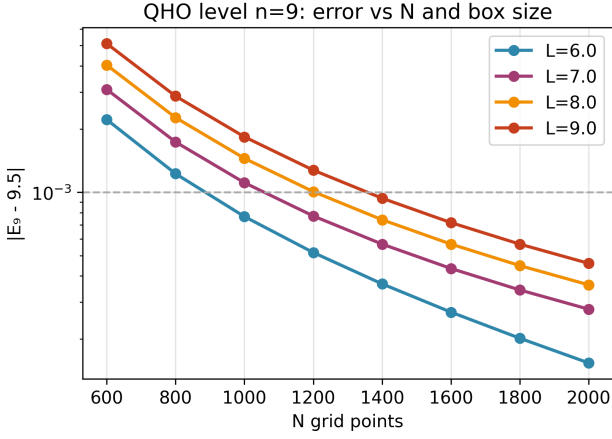


Figure 1: Energy error for the highest resolved QHO state ($n = 9$) plotted as $|E_9 - 9.5|$ against the number of grid points N for several choices of box half-width L . As the resolution increases, the finite-difference truncation error decreases, while enlarging L suppresses boundary effects. For $L \gtrsim 8$ the curves nearly coincide, indicating that box error is negligible and the remaining behaviour follows the expected N^{-2} discretisation scaling.

ensure that box truncation is already secondary to the finite-difference error. Beyond this regime, increasing L or N yields only minor accuracy gains while incurring additional computational cost.

2.1.2 Baseline eigenvalues and eigenfunctions

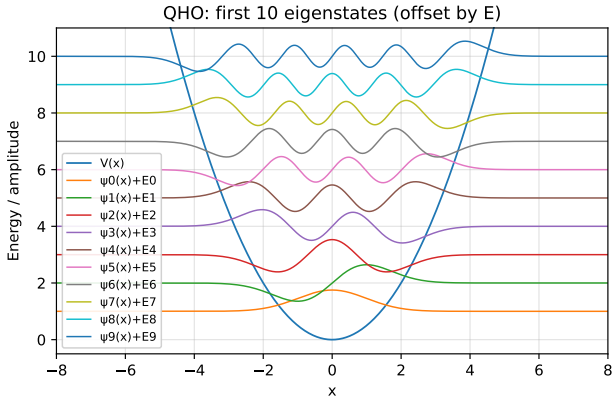


Figure 2: Harmonic-oscillator potential $V(x) = \frac{1}{2}x^2$ and the first ten eigenstates, each vertically shifted by its eigenvalue E_n . The numerical wavefunctions exhibit the correct parity alternation and node structure, with the n th state containing n zeros, confirming that the chosen discretisation resolves the expected qualitative features.

The computed eigenfunctions display the expected qualitative features of the QHO: the correct alternation of parity, the appropriate number of nodes, and the anticipated spatial confinement [3]. Their qualitative agreement with the analytic shapes is clear. The maximum energy deviation remains within the prescribed tolerance ($\max |\Delta E| \approx 7.4 \times 10^{-4}$), and the wavefunction values at the domain edges fall between 10^{-12} and 10^{-6} , indicating that the states are effectively negligible at the boundaries.

2.1.3 Higher-order finite differences and convergence

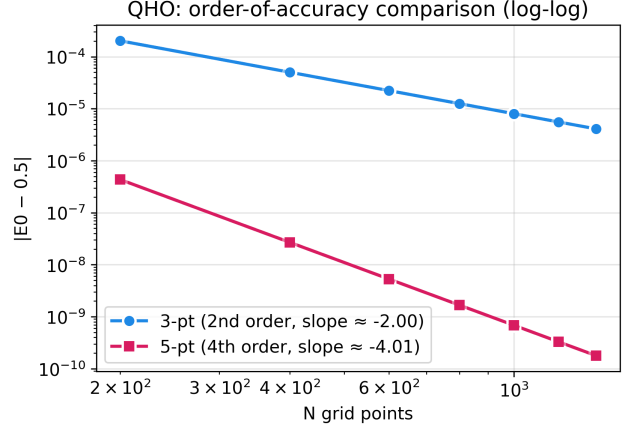


Figure 3: QHO: log-log convergence test for the three-point (second-order) and five-point (fourth-order) finite-difference stencils. The fitted slopes, $p \approx 2$ and $p \approx 4$, confirm the expected truncation orders and show that the five-point scheme attains higher accuracy than the three-point method for the same grid size N .

Since the order of accuracy comparison shows the five-point stencil attaining notably smaller errors for similar runtimes, we repeated the QHO calculation using the fourth-order scheme to evaluate its effect on the final results. The eigenfunctions themselves change very little: their shapes are visually identical, and the boundary amplitudes remain between 10^{-6} and 10^{-12} , confirming that domain truncation is already negligible. The energy errors, however, improve markedly. For $L = 8$ and $N = 1400$, the maximum deviation decreases from 7.4×10^{-4} with the three-point stencil to 2.1×10^{-7} with the five-point version. Thus, in the discretisation-dominated regime, the five-point method delivers substantially greater accuracy, although the three-point stencil still meets the tolerance of Section 1.3 and remains the more economical choice [4].

2.2 Hydrogen radial equation ($\ell = 0$)

2.2.1 Domain and grid parameter calibration

We determine $(R_{\max}, N)$ from the ground state alone: $R_{\max}$ is chosen from the region of Fig. 4 where the error becomes insensitive to further increases in box size, and N is then selected so that $|E_{1,\text{num}} + \frac{1}{2}|$ meets the hydrogen tolerance specified in Section 1.3. The final choice $R_{\max} = 60$ and $N = 1600$ yields a ground-state error well below 10^{-3} with negligible boundary effects. This grid is used for all remaining hydrogen calculations.

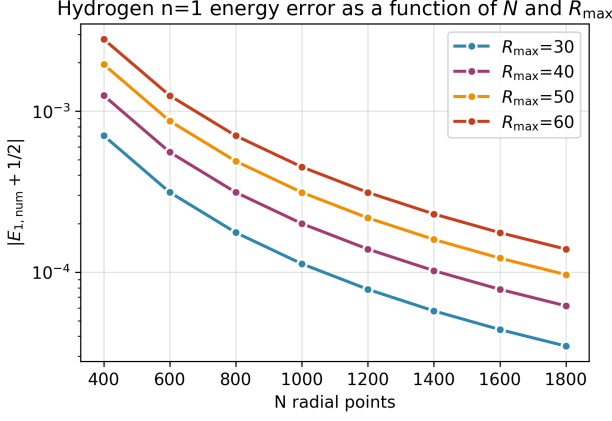


Figure 4: Hydrogen $\ell = 0$, ground state $n = 1$: energy error $|E_{1,\text{num}} + \frac{1}{2}|$ plotted against the number of radial grid points N for several choices of the domain cutoff R_{max} . Increasing N reduces the finite-difference discretisation error with its characteristic N^{-2} scaling, while enlarging R_{max} suppresses boundary effects. For $R_{\text{max}} \gtrsim 60$, the curves collapse onto a common N^{-2} trend, showing that box error is no longer the dominant contribution.

2.2.2 Radial eigenvalues and eigenfunctions

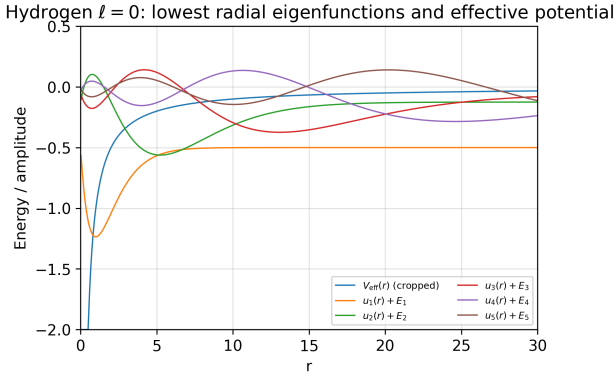


Figure 5: Hydrogen $\ell = 0$: effective radial potential $V_{\text{eff}}(r)$ together with the lowest five eigenfunctions $u_n(r)$, computed on the calibrated grid ($R_{\text{max}} = 60$, $N = 1600$). Each curve is shifted upward by its eigenvalue E_n . The wavefunctions are regular at $r = 0$, exhibit the correct increase in node number with n , and remain well confined within the chosen domain.

Using this grid, we compute the first five $\ell = 0$ energy levels. Their numerical eigenvalues satisfy $|E_{n,\text{num}} - E_n| \leq 4 \times 10^{-4}$ for $1 \leq n \leq 5$, comfortably within the hydrogen tolerance of Section 1.3. The associated eigenfunctions shown in Fig. 5 exhibit the expected analytic features; regular behaviour $u_n(r) \propto r$ near the origin, $n - 1$ nodes, and rapid decay for $r \gtrsim 20$. This confirms that the discretised radial Hamiltonian accurately reproduces the bound-state structure before we move on to algorithmic comparisons [3].

2.2.3 Numerov vs finite-difference solver

Figures 6 and 7 demonstrate that the Numerov method and the three-point FD + `eigsh` solver behave in line

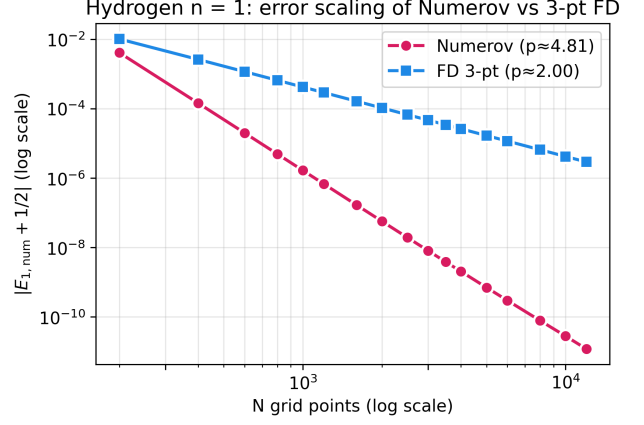


Figure 6: Hydrogen $n = 1$: log-log plot of the energy error $|E_{1,\text{num}} + \frac{1}{2}|$ versus grid size N for the Numerov method and the three-point FD + `eigsh` solver. The fitted slopes, $p_{\text{Num}} \approx 4.9$ and $p_{\text{FD}} \approx 2.0$, reflect the characteristic sixth- and second-order convergence of the two schemes once pre-asymptotic behaviour has subsided.

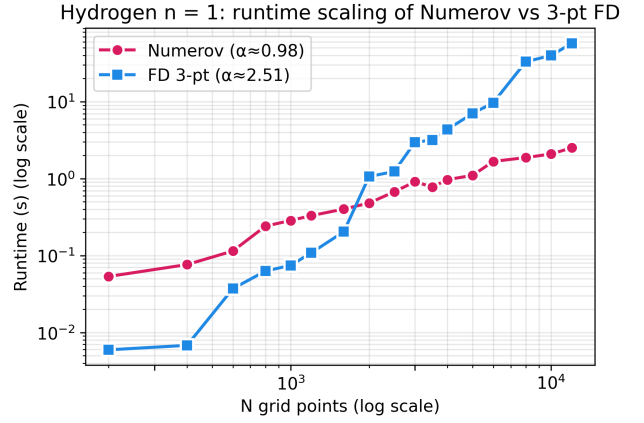


Figure 7: Hydrogen $n = 1$: log-log runtime scaling for the Numerov method and the three-point FD + `eigsh` solver. Numerov shows an almost linear growth with grid size ($\alpha_{\text{Num}} \approx 0.98$), while the FD approach scales more steeply ($\alpha_{\text{FD}} \approx 2.51$), characteristic of sparse iterative eigensolvers.

with their expected accuracy orders. Numerov exhibits a markedly sharper error decrease ($p \approx 4.9$) than the finite-difference approach ($p \approx 2.0$), giving errors several orders of magnitude smaller at the same grid sizes. Yet this improvement comes with different runtime characteristics. The scaling exponents show that Numerov grows almost linearly with grid size ($\alpha \approx 0.98$), whereas the FD + `eigsh` method grows more quickly ($\alpha \approx 2.51$). For the problem sizes of interest here ($N \lesssim 2 \times 10^3$), the finite-difference solver is still the faster option because a single sparse eigenvalue computation carries a smaller prefactor than Numerov's repeated ODE integrations and bisection steps. The two curves meet only near $N \approx 2 \times 10^3$, a resolution well above what is required to meet our accuracy target, so Numerov becomes preferable only when substantially higher precision or much larger domains are required.

Conclusion

The QHO and hydrogen solvers were tuned to the $\sim 10^{-3}$ accuracy target, with computed spectra agreeing with analytic levels well within their natural spacings. After boundary effects were reduced and discretisation error became dominant, we identified $N \lesssim 2 \times 10^3$ as the range where the error saturates: beyond this point the boundary is no longer the limiter, and increasing N produces only marginal gains. Within this regime the three-point FD method is the most efficient. The five-point stencil lowers truncation error, but once the plateau is reached its extra bandwidth adds cost without improving the tolerance. Numerov shows its expected higher-order convergence only at much tighter targets ($\ll 10^{-3}$) or for significantly larger grids ($N \gtrsim 4 \times 10^3$), consistent with its asymptotic behaviour.

Beyond this assignment, the same FD and Numerov tools underpin the numerical analysis within our Riemann Hypothesis project, where we compute Laplace–Beltrami spectra of negatively curved surfaces to compare their statistics with those of the Riemann zeros.

Use of Generative AI

I used ChatGPT 5.1 as a general aid while working on this assignment. Its role was mainly to provide guidance on implementation details, such as how to structure the Numerov method, construct the fourth-order finite-difference stencil, and organise the sparse eigensolver workflow. All code was written and adapted by me; the AI was used only for suggestions and clarification of numerical methods. Instances where AI input informed a specific coding choice are explicitly noted in the corresponding code cells.

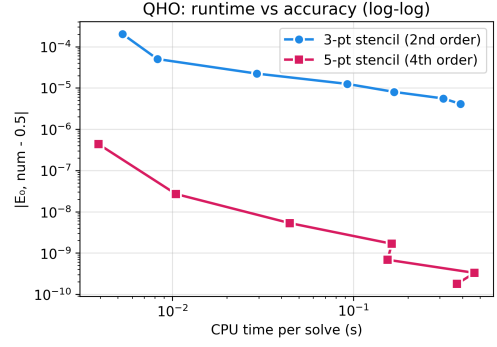


Figure 9: QHO: log–log comparison of runtime versus ground–state error for the three–point and five–point finite–difference stencils. The fourth–order scheme achieves substantially lower truncation error for a given accuracy, but at a higher computational cost per solve. Small irregularities in the curves arise from timing variability rather than differences in algorithmic behaviour.

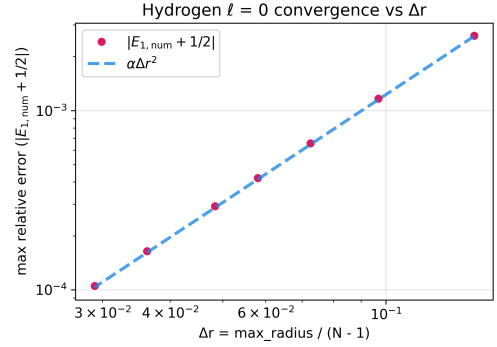


Figure 10: Hydrogen $\ell = 0$, first bound state: log–log plot of the relative energy error $|E_{1,\text{num}} + \frac{1}{2}|$ as a function of the radial spacing Δr for the three–point finite–difference scheme. The observed slope of approximately 2.00 verifies the expected second–order convergence of the radial discretisation.

Appendix: Additional diagnostics

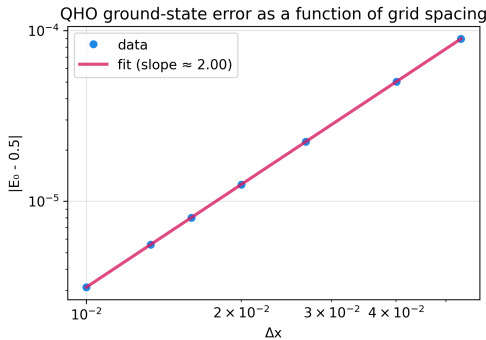


Figure 8: QHO ground state: log–log plot of the absolute error $|E_{0,\text{num}} - 0.5|$ as a function of grid spacing Δx computed with the three–point finite–difference stencil. The fitted line has slope ≈ 2.00 , demonstrating the expected second–order convergence of this discretisation.

References

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- [2] P. Virtanen, R. Gommers, T. E. Oliphant *et al.*, “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python,” *Nature Methods* **17**, 261–272 (2020).
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